

Sole Source Justification and Approval
Under FAR 13.501

1. Contracting Activity: U.S. Army Medical Research Acquisition Activity (USAMRAA), 808 Shreider Street, Fort Detrick, MD 21702-5014

2. Description of Action: The U.S. Army Medical Research Institute of Infectious Diseases (USAMRIID) requires a new, sole source, Firm Fixed Price (FFP) contract with Aptagen, LLC located at 250 N Main Street, Jacobus, PA 17407. The resultant contract will be funded utilizing FY25 Defense Health Program Research, Development, Test and Evaluation (RDT&E) funding.

3. Description of Supplies/Services: USAMRIID has a need to generate two (2) Deoxyribonucleic Acid (DNA) Aptamer Library Generation per anticipated projects per year. These aptamer libraries will consist of ~1000 - ~10,000 targeted aptamers. These libraries will target specific alpha-conotoxins and palytoxins and should be capable of broad-spectrum interactions and libraries targeting four (4) to six (6) viral targets. This is a new requirement that was identified to support USAMRIID's project [REDACTED], Development of broad-spectrum DNA aptamers to neutralize toxins. The requirement is crucial as it will enable USAMRIID's researchers to ultimately generate one (1) or more DNA aptamers capable of effectively binding and neutralizing specific toxins and viruses. This is an embryonic methodology with promising outlook for future use in the research field as this and similar strategies have not been employed at the institute previously. Additionally, the institute lacks personnel with the level of technical knowledge and resources required to develop these specific libraries, requiring USAMRIID to seek support from developers external to the institute.

[REDACTED] This value is inclusive of one (1) twelve-month base Period of Performance (PoP), two (2) twelve-month option periods and one (1) optional six (6) month extension in accordance with FAR 52.217-8 Option to Extend Services.

4. Authority Cited: 41 United States Code 1901 as implemented in Federal Acquisition Regulation 13.501(a)(1)(ii) sole source (including brand name) under the authority of simplified procedures for certain commercial items.

5. Reason for Authority Cited: Market research was conducted in September and October 2024. During this research a total of three (3) sources were researched and

identified as potential sources. Due to the specialized nature of our research question, and our use of select agents, only one (1) of the potential sources emerged as a viable option capable of meeting the Government's requirements. USAMRIID requires a specialized knowledge set and corresponding supplies to generate DNA aptamer libraries (small DNA molecules that interact specifically with a given molecule), which is only offered by a small number of contractors, along with the ability to generate these libraries in containment. Aptagen LLC is the only vendor capable of providing reagents that can be used on site to initiate the process, train in their use, and then synthesize and characterize the aptamers that are promising candidates. Aptagen, LLC is a leader in aptamer library generation. Aptagen owns its proprietary processes of aptamer development. They are the only company to provide aptamer library kits along with library characterization services. Further, Aptagen LLC provides the ability for USAMRIID to leverage its expertise while retaining possession of the select agent in accordance with USAMRIID's strict and rigorous access and control protocols, allowing for the production of the libraries within said containment parameters.

6. Efforts to Obtain Competition: In accordance with FAR 5.2 - Synopses of Proposed Contract Actions, a Notice of Intent (NOI) to award a sole source will be posted to www.sam.gov for a period of 15 days. All capability statements received in response to the NOI to Sole Source, by the due date specified, will be considered by the Government solely for the purpose of determining whether to conduct a competitive procurement or to proceed with a sole source contract.

7. Actions to Increase Competition: USAMRIID and USAMRAA will periodically conduct future market research and analysis to determine if other potential sources emerge that can provide the required aptamer library generation. Additionally, all future library generation requirements will be synopsized in accordance with FAR 5.201 and if other potential sources do emerge, USAMRAA in conjunction with USAMRIID will assess whether competition for future requirements is feasible.

8. Market Research: Market Research conducted in accordance with FAR Part 10 was multifaceted and was undertaken to ascertain 1) if the requirement meets the FAR definition of Commercial Product/Commercial Service; 2) if two or more small businesses exist that can meet the Government's requirements; and 3) if there is more than one source capable of meeting the governments requirements.

Market Research examined if the required service met the FAR definition of a commercial services, which is services of a type offered and sold competitively in substantial quantities in the commercial marketplace. Market Research determined that the required service does meet this definition and is commercially available; thus, standard commercial terms and conditions will apply.

Market research conducted in accordance with FAR Part 10 does not support that a minimum of two or more capable small businesses could potentially satisfy this requirement. Therefore, this requirement is not appropriate for a small business set-aside.

USAMRIID is aware that the marketplace for such services may be limited by the number of companies that could potentially respond to this requirement. USAMRIID is not aware of any laws or regulations unique to this service. They are not aware of available services that require recovered materials and that are energy efficient and there are no accommodations that need to be considered for entry into the service location. Based on our assessment the typical standard service timeframe for this requirement is two (2) to three (3) years. We believe there is a medium demand for this service as it is a foundational stage to the research and development of compounds essential in the fight against known toxins and diseases.

The market research team performed multiple internet searches seeking to determine what companies have the capabilities required to fill the needs of this requirement. Of the available companies the above listed three (3) sources were either contacted or researched (AMSBIO, LLC, Aptagen, LLC, and Creative Biolabs). Although all three (3) can fulfill certain aspects of this requirement, Aptagen, LLC emerged as the only source capable of providing all USAMRIID requirements. Aptagen LLC is the only identified source capable of providing kits and training to perform library generation within the constraints of USAMRIID's containment protocols. This meets USAMRIID's mandate to maintain targets on-site and in containment. Additionally, Aptagen can perform other portions of aptamer library generation, such as aptamer synthesis and candidate selection that do not involve direct use of the target toxin. AMSBIO LLC and Creative Biolabs cannot perform these tasks. This enables us to leverage their knowledge and expertise without the potential risk from removing select agents from the institute.

9. Interested Sources: To date, no sources have written to express an interest in this requirement. The synopsis required by FAR 5.201 will be posted to www.sam.gov and will allow all interested sources to respond with a Capability Statement. Any responses received will be considered by the Government solely for the purpose of determining whether to conduct a competitive procurement or to proceed with a sole source.

10. Other Facts:

- a. Procurement history. Not applicable – This is a new requirement.
- b. Other Facts. Not applicable.